

Non-silicone elastic coating with fouling resistance and fouling release abilities based on degradable hyperbranched polymer

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ABSTRACT

Developing eco-friendly antifouling materials for combating marine biofouling has long been a challenging task. Herein, we report a non-silicone elastomer coating based on hyperbranched poly(ϵ -caprolactone) (HPCL) cross-linked with an amphiphilic triblock copolymer of poly(ethylene glycol)-*block*-poly(propylene glycol)-*block*-poly(ethylene glycol). The polymer coating is prepared by ring-opening polymerization and self-condensation reactions, which can be prepared at a large scale. Such a coating is expected to have a self-renewed surface, fouling resistance and fouling release abilities. Quartz crystal microbalance with dissipation test shows that the coating is degradable, leading to a self-renewed surface. The degradation rate can be regulated by adjusting the HPCL content. Bioassays with bacteria (*Staphylococcus aureus* and *Escherichia coli*) and diatom (*N. incerta*) reveal that the surface with triblock copolymer content of 12.5 wt% or higher can resist the adhesion of microorganisms. The coating also has good fouling release performance with pseudobarnacle removal strength of ~ 0.15 MPa, which is comparable with that of silicone elastomer. Marine field test for over 120 days proves the coating is effective in inhibiting marine biofouling due to the synergetic effect of surface renewal, fouling release and fouling resistance. This work provides a facile preparation method for biocide-free antifouling materials.

1. Introduction

Marine biofouling is the undesired settlement and accumulation of marine organisms on materials submerged in seawater [1,2]. This increasingly serious problem has detrimental effects on maritime transportation, marine energy facilities, and marine aquaculture industries [3–5]. In recent years, coatings based on metal biocides (mostly cuprous oxide) and self-polishing copolymers are widely used for marine antibiofouling. However, these coatings are not effective under static conditions because the surface renewal depends on the strong flow of water, and the release of massive metal biocides and non-degradable polymer binder causes ecological hazards to the marine system [6,7]. Hence, eco-friendly antifouling technologies have emerged, including fouling release coatings (FRCs) [8–10], degradable polymers [11,12], bioinspired antifouling materials [13,14], amphiphilic polymers [15,16], and protein resistant polymers [17–19]. Among them, FRCs without any biocides have received increasing attention because of their excellent fouling release ability and environmental friendliness.

High elasticity (low modulus), low surface energy and smooth surface are keys to weakening the adhesion of marine biofouling on the

coatings and promoting the removal of accumulated organisms. In particular, an elastic surface can reduce the adhesion of organisms because of its instability for adhering [20,21]. Conventionally, silicone elastomers possessing these characteristics are used for preparing FRCs. However, they have limited antifouling efficacy under static conditions because the fouling release performance depends strongly on strong water flow, and they do not contain antifouling group [22]. Moreover, they have poor mechanical durability and weak adhesion to substrates [23–25]. Therefore, developing new FRCs materials besides silicone elastomers is an important direction.

Meanwhile, to meet the complex marine environments and diversified fouling species, fouling release can be combined with other properties that are beneficial for anti-biofouling. For example, amphiphilic polymers composed of both hydrophobic and hydrophilic components can effectively resist fouling organisms [26,27]. Combining the fouling resistance of amphiphilic polymer with the fouling release of FRCs is a promising strategy, which can simultaneously prevent the initial settlement of organisms and weaken the adhesion strength of attached ones. Moreover, a self-renewing dynamic surface consisting of degradable polymers has proved effective in inhibiting biofouling. Since their

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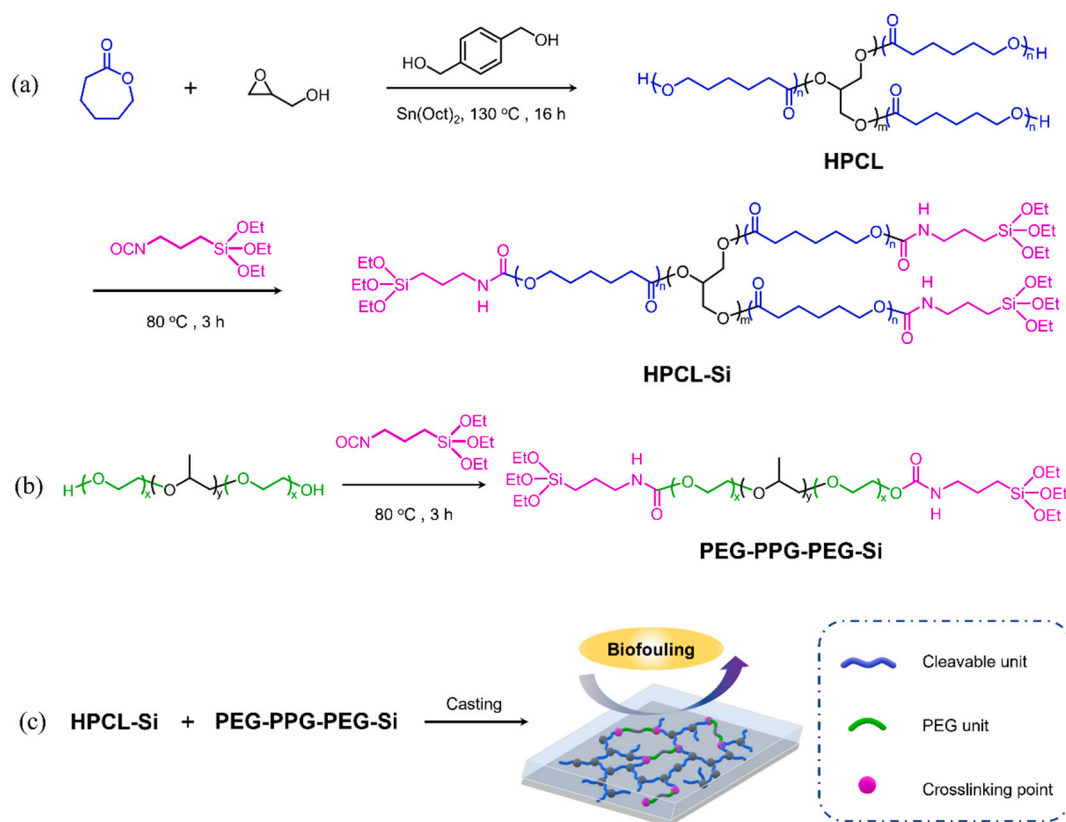
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Scheme 1. Schematic diagram of the synthesis of HPCL-Si (a) and PEG-PPG-PEG-Si (b) and HPCL-Tx (c)

Table 1
Characterization data of HPCL-Tx.

Sample	CL (wt %)	PEG-PPG-PEG-Si (wt %)	T_g ($^\circ\text{C}$) ^a	T_m ($^\circ\text{C}$) ^a	ΔH_m (J/g) ^a	X_c (%) ^b
HPCL-T0	63.5	0.0	-53.3	27.8	36.1	41.8
HPCL-T12.5	55.6	12.5	-56.3	22.7	27.3	36.1
HPCL-T25	47.6	25.0	-58.5	19.9	8.8	13.6
HPCL-T37.5	39.7	37.5	-60.9	11.1	4.6	8.52
HPCL-T50	31.8	50.0	-62.7	5.6	3.5	8.09

^a Determined by DSC.

^b Calculated from $X_c(\%) = \Delta H_m / w \Delta H_0 \times 100$, where w represents the weight proportion of PCL segments in the network and ΔH_0 for 100 % crystalline PCL is 136 J/g [30].

main-chain degradation does not rely on strong water flow, they have excellent efficacy under static conditions [28,29]. Therefore, constructing an antifouling system with fouling release and fouling resistance with degradable polymer rather than silicone may have remarkable performance.

Existing antifouling materials in use, both biocide-based coatings and FRCs, usually have limited antifouling efficacy under static condition and have only a single antifouling function. Constructing a multifunctional antifouling coating can be a potential strategy to combat marine biofouling. Herein, we prepared a non-silicone coating based on degradable hyperbranched poly(ϵ -caprolactone) (HPCL) cross-linked with an amphiphilic triblock copolymer PEG-PPG-PEG. The polymer coating was prepared using a facile procedure of ring-opening polymerization and self-condensation reactions, which can be prepared at a large scale. The PEG-PPG-PEG amphiphilic additive can provide resistance to protein and microorganisms, and the cross-linked structure

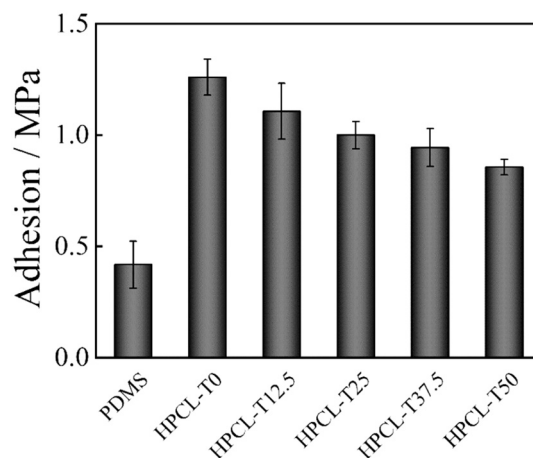


Fig. 1. Adhesion strength of HPCL-Tx coatings and PDMS reference on the substrate (epoxy panel).

leads to a highly elastic coating. In addition, the HPCL is expected to degrade into fragments to form self-renewing dynamic surfaces. In other words, the coating can achieve fouling resistance, fouling release properties and self-renewing surfaces at the same time. The mechanical properties, degradation, fouling resistance, and fouling release abilities of the prepared elastomer have been examined. This study provides an approach for the development of a multifunctional and synergistic antifouling coating for marine applications.

2. Results and discussion

We have prepared a non-silicone elastomer coating based on HPCL

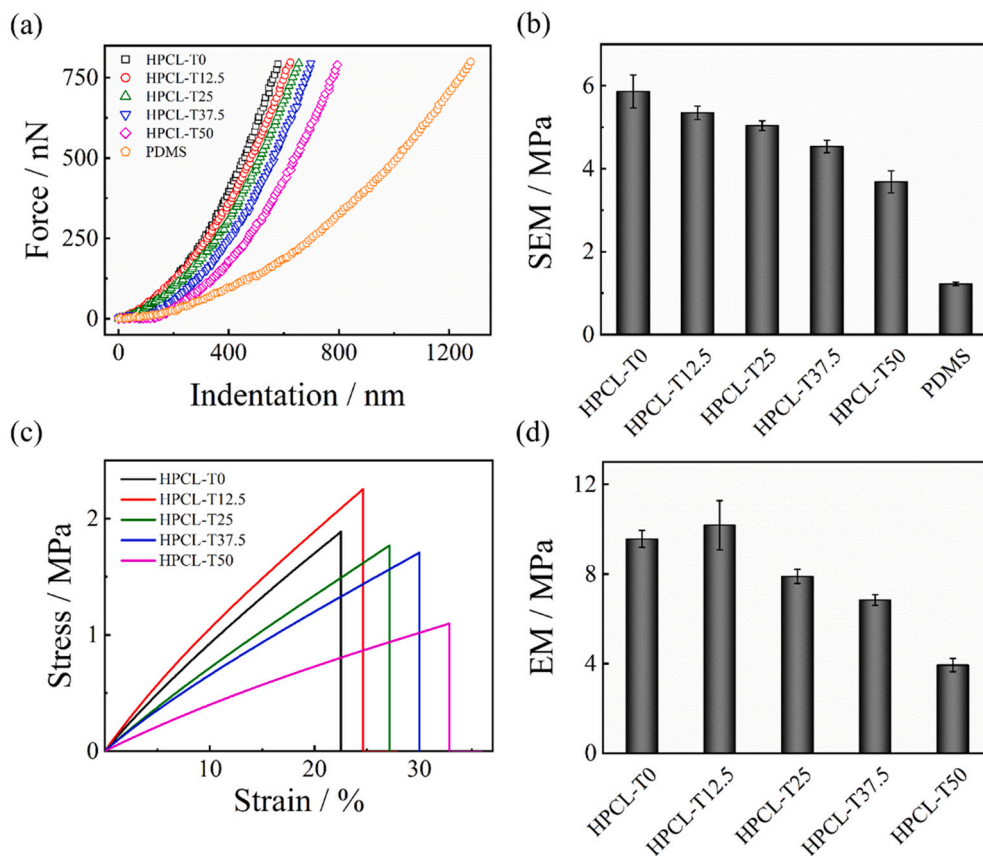


Fig. 2. Force-indentation curve (a) and surface elastic modulus (SEM) (b) of HPCL-Tx coatings at 25 °C from AFM. Stress-strain curves (c) and elastic modulus (EM) (d) of HPCL-Tx coatings at 25 °C from tensile test.

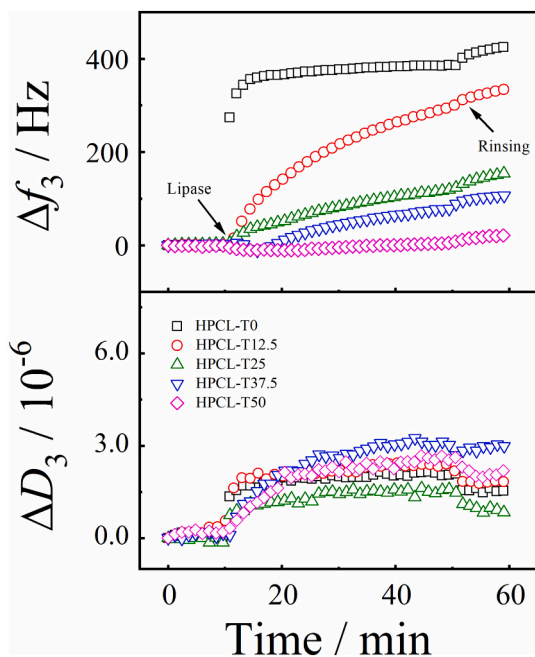


Fig. 3. Time dependences of frequency shift (Δf) and dissipation shift (ΔD) for the enzymatic degradation of HPCL-Tx coating as a function of PEG-PPG-PEG content in ASW at 25 °C.

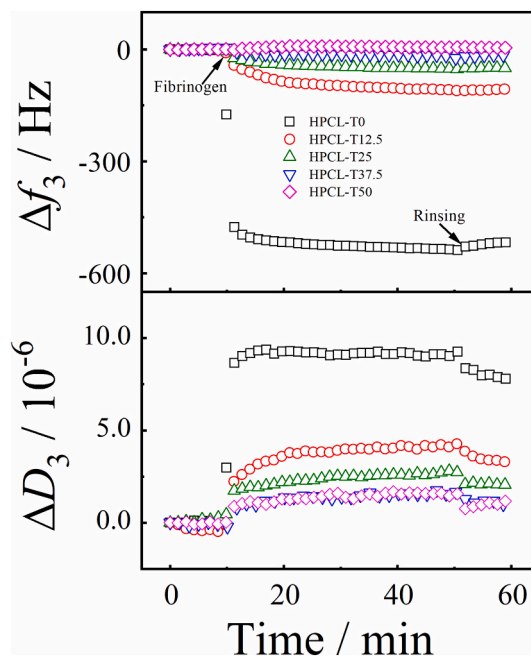


Fig. 4. Change of frequency (Δf) and dissipation (ΔD) regarding adsorption of fibrinogen on HPCL-Tx coatings in ASW at 25 °C.

cross-linked with an amphiphilic triblock copolymer. As illustrated in Scheme 1, HPCL was synthesized by ring-opening polymerization of CL

and then modified by silane (HPCL-Si), triblock copolymer PEG-PPG-PEG was also modified with silane (PEG-PPG-PEG-Si), and the film was prepared by self-condensing of HPCL-Si and PEG-PPG-PEG-Si.

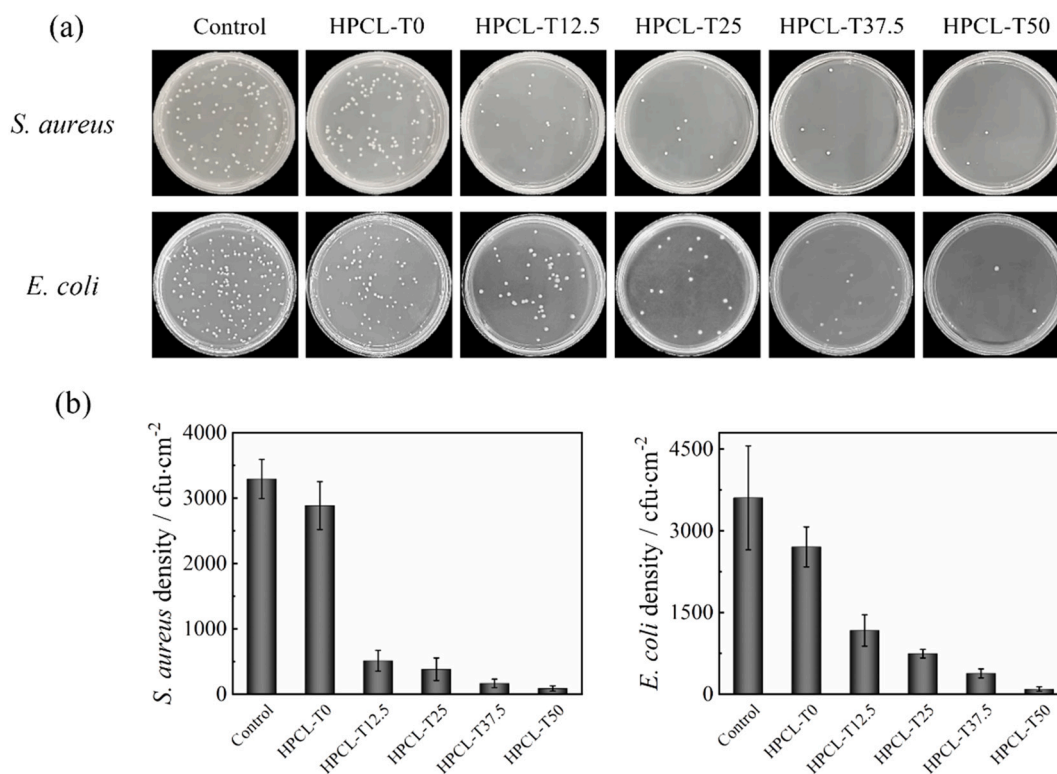


Fig. 5. Colonial morphology images (a) and the numbers of bacteria *S. aureus* and *E. coli* adhered on the HPCL-Tx coating surfaces (b).

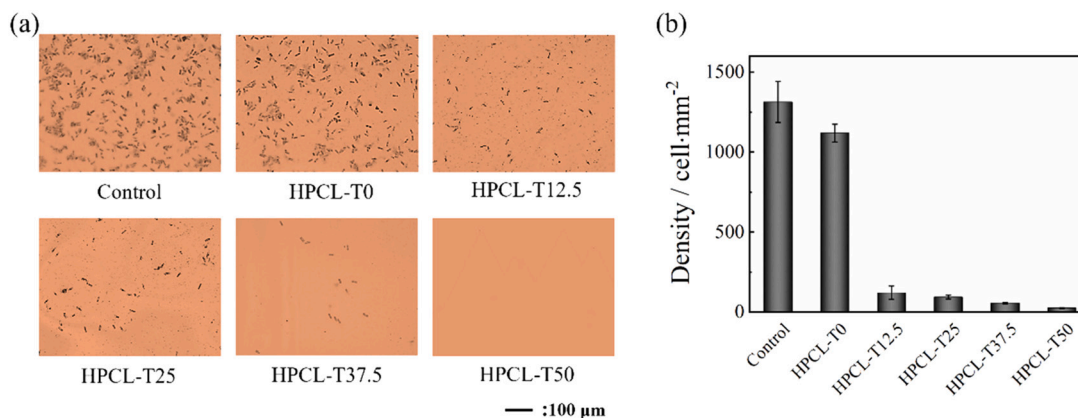


Fig. 6. Optical microscopy images of the diatom *Navicula incerta* adhered to HPCL-Tx (a) and the quantitative result (b).

Coatings with different compositions were prepared to study the influence of the component content on the performance. The coating is named as HPCL-Tx, where x represents the PEG-PPG-PEG-Si content. More characterization data of HPCL-Tx coating can be found in Table 1.

The coating is anticipated to have excellent fouling resistance and fouling release properties. Moreover, it can form self-renewing dynamic surfaces. The methodology used to characterize the material mainly focus on the antifouling and mechanical properties. In particular, not only lab assays were conducted but marine field test was used to assess the long-term antifouling performance in nature seawater. Marine field results correspond well with laboratory results.

Fig. 1 shows the adhesion strength of HPCL-Tx on the epoxy substrate. PDMS elastomer, which is widely used for FRCs, was also examined. PDMS shows a low adhesion of 0.4 MPa because it has a nonpolar structure, causing the coating readily detach from the substrate during application. By contrast, the HPCL-Tx coatings exhibit a relatively high adhesion strength. HPCL-T0 has the highest adhesion

strength of 1.3 MPa because of its polar ester bonds. The adhesion strength of HPCL-T12.5 slightly decreases (1.1 MPa) because the polarity of $-C-O-C-$ in PEG-PPG-PEG triblock copolymer is lower than that of the ester bond in HPCL-Si, thereby weakening the interaction between coating and substrate. When the PEG-PPG-PEG content increases to 50 wt%, the adhesion strength gradually decreases to 0.9 MPa. On the one hand, the increase in $-C-O-C-$ content reduces the adhesion strength. On the other hand, the T_g of HPCL-Tx decreases from -53.3 to -62.7 °C, thus leading to a reduction in mechanical strength and adhesion strength.

Fig. 2a, b shows the AFM force-indentation curve and surface elastic modulus of the HPCL-Tx coatings. At AFM force signal (F) of 800 nN, PDMS has the largest indentation depth and therefore the largest deformation, implying its good elasticity. Meanwhile, HPCL-T0 has the smallest indentation depth and deformation because it slightly crystallizes and is rigid. The indentation depth increases with the PEG-PPG-PEG content, which shows an improvement in elasticity. In addition, the

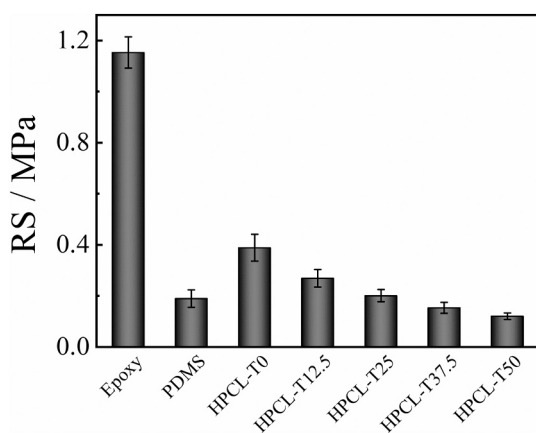


Fig. 7. Removal strength (RS) of pseudobarnacles HPCL-Tx coatings and PDMS reference.

curve of the HPCL-Tx coating gradually approaches the PDMS curve, indicating that the elasticity is close to that of PDMS. Fig. 2b shows the SEM of HPCL-Tx coatings. PDMS has the smallest SEM of 1.2 MPa. Whereas HPCL-T0 appears the largest SEM of 5.9 MPa. With a further increase in PEG-PPG-PEG content, the SEM gradually decreases to 3.7 MPa, demonstrating the enhancement of elasticity. In this work, when PEG-PPG-PEG is introduced, the T_g of films decreases (Table 1), and their elasticity further increases after cross-linking.

Tensile test was also conducted to examine the mechanical properties of the HPCL-Tx coatings (Fig. 2c, d and Table S1). Compared to HPCL-T0 with 1.8 MPa. HPCL-T12.5 has a slightly higher tensile strength of 2.2 MPa because the introduction of PEG-PPG-PEG increases the cross-linking density. However, further increase in the PEG-PPG-PEG content gradually decreases the strength, because the triblock copolymer softens the polymer as reflected by the decreased T_g . The trend of elastic modulus of HPCL-Tx is similar to that of tensile strength. It is worth noticing that all the polymers have low elastic modulus ranging from 10.2 to 3.9 MPa. AFM and tensile test results reveal that the polymers have high elasticity and low modulus, which can contribute to the fouling release property.

Many microorganisms in marine environments can excrete enzymes. Biodegradable polymers can degrade under the attack of enzymes [31]. Therefore, we first used QCM-D to rapidly assess the enzymatic degradation of the polymers. The lipase PS was chosen to evaluate the enzymatic degradation performance of HPCL-Tx. Fig. 3 shows the time dependence of frequency shift (Δf) and dissipation shift (ΔD) of the enzymatic degradation of HPCL-Tx in ASW. It is well known that the increase in frequency is due to a decreased mass of the sensor surface [32]. The Δf of HPCL-T0 with the highest CL content remarkably

increases relative to the baseline after introducing lipase PS, and afterwards it stabilizes. After rinsing by ASW, there is still a slight increase in Δf , proving the decreased mass of the film. For HPCL-T12.5, Δf increases relative to the baseline, but is less than that for HPCL-T0 because of the reduced CL contents. For HPCL-T25, HPCL-T37.5, and HPCL-T50 with low CL contents, Δf slightly increases relative to the baseline after rinsing. It can be seen from the high increase in Δf that the degradation rate increases with CL content because polyester chains are degraded into small fragments dispersing in solution. ΔD reflects the structural changes of the film, which correlates with the viscoelasticity of the additional layer. A dense and rigid film on the surface of QCM-D sensor results in a low ΔD , while a loose structure leads to large dissipation [33,34]. Compared with the baseline, HPCL-T0 has a slight increase in ΔD , indicating that the film becomes thin due to its high degradation rate. For HPCL-T12.5, the increase in ΔD is due to its high degradation rate and rough surface. As the PEG-PPG-PEG content is increased to 50 wt%, the change in ΔD gradually decreases but its value remains higher than that in HPCL-T0. This phenomenon occurs because when the content of CL segments decreases, the degradation rate also decreases and the film structural changes become relatively less. Moreover, the crystallinity of polyester affects its enzymatic degradation to some extent. As shown in Table 1, with the increase in PEG-PPG-PEG content, the crystallinity (X_c) decreases. Thus, the introduction of PEG-PPG-PEG makes the degradation heterogeneous, resulting in an uneven surface with a large ΔD . In brief, the QCM-D test shows all the samples are capable of degradation.

Protein adsorption is generally considered the first step in biofouling formation because many fouling organisms adhere on submerged objects by secreting protein-based adhesive [35]. The protein resistance on HPCL-Tx coatings was examined using QCM-D. Fibrinogen was selected as the model protein. Fig. 4 shows the time dependences of frequency shift (Δf) and dissipation shift (ΔD) of fibrinogen adsorption on the films. HPCL-T0 shows a significantly decreased Δf and increased ΔD after ASW rinsing, demonstrating a large amount of protein adsorbs on its surface. HPCL-T12.5 exhibits a smaller change in Δf and ΔD , indicating that fewer protein adsorbs on the surface. With increasing PEG-PPG-PEG content, Δf is greatly reduced until its final value is almost unchanged relative to the baseline. A similar change is observed in ΔD . In particular, HPCL-T37.5 and HPCL-T50 exhibit significant protein resistance, their Δf and ΔD return to baseline after rinsing. As the PEG-PPG-PEG content increases, the water contact angle of the coating gradually decreases, that is, the surface hydrophilicity also increases (Fig. S1). This is because the hydration effect of the coating surface is strengthened due to the PEG segment, which is beneficial for protein resistance [36–38].

The adhesion of bacteria is a key step in the formation of biofilm and affects the colonization of macrofouling organisms [39]. The anti-adhesion ability of HPCL-Tx was assessed by two types of bacteria,

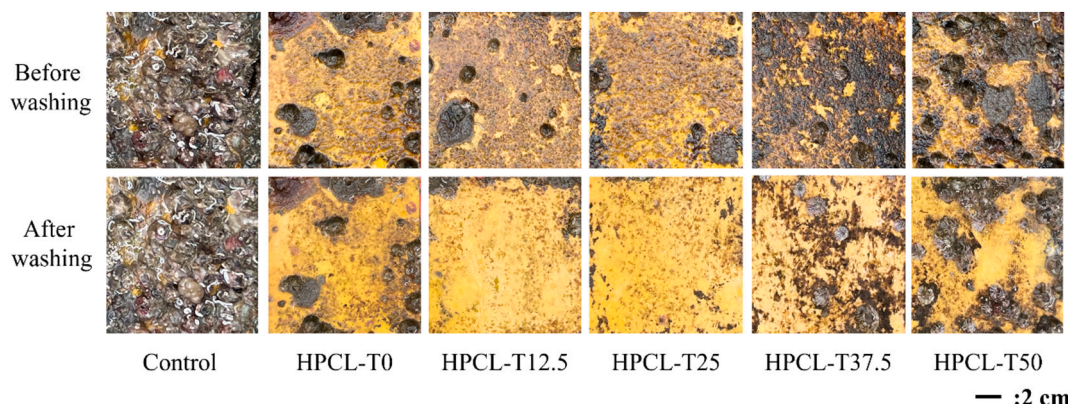


Fig. 8. Images of HPCL-Tx coatings after immersion in the South China Sea for 120 days, before and after a water jet treatment.

S. aureus (Gram-positive) and *E. coli* (Gram-negative). Fig. 5 shows high-density *S. aureus* bacterial colonies on the blank control coverslip ($3300 \text{ cfu}\cdot\text{cm}^{-2}$) and HPCL-T0 coating ($2900 \text{ cfu}\cdot\text{cm}^{-2}$) after 4 h of exposure to bacterial suspension. By comparison, only a few bacterial colonies are found on the samples with PEG-PPG-PEG, indicating that the amphiphilic additive inhibits bacterial adhesion because of the hydration layer on the surface of the coating. With increasing PEG-PPG-PEG content, the antibacterial efficiency is improved with density values of $150 \text{ cfu}\cdot\text{cm}^{-2}$ and $80 \text{ cfu}\cdot\text{cm}^{-2}$ for HPCL-T37.5 and HPCL-T50, respectively. That is, the coating containing PEG-PPG-PEG shows better resistance to bacteria than HPCL-T0. Namely, the reductions of bacterial densities of HPCL-T37.5 and HPCL-T50 are both higher than 95 % compared with the blank control. The results for *E. coli* are similar to those for *S. aureus* with only slight difference in bacteria density values, thus implying the excellent antibacterial property of the HPCL-Tx coatings.

Fig. 6a, b demonstrates the images of HPCL-Tx surface after immersing in diatom *N. incerta* and the quantitative density of adhered diatom. A great number of diatoms have colonized on the blank control silicon wafer and HPCL-T0 surfaces ($1300, 1100 \text{ cell}\cdot\text{mm}^{-2}$) in that diatoms are prone to adhere on hydrophobic surfaces [23]. By comparison, the diatom densities on the coatings with PEG-PPG-PEG are relatively low at 120, 90, 50, and $20 \text{ cell}\cdot\text{mm}^{-2}$ on HPCL-T12.5, HPCL-T25, HPCL-T37.5, and HPCL-T50 surfaces, respectively. As discussed above, the hydration layer contributes significantly to the anti-diatom performance. Compared with unmodified silicone coating [22], the HPCL-Tx coatings have better resistance to diatom. This result correlates well with those from protein and bacteria tests, proving that the HPCL-Tx coatings with PEG-PPG-PEG content of 12.5 wt% or higher have excellent fouling resistance.

Fig. 7 shows the removal strength (RS) of HPCL-Tx coating measured by the pseudobarnacle removal test with PDMS as a reference material, reflecting the fouling release property. Clearly, the removal strength of epoxy is $\sim 1.1 \text{ MPa}$, which clearly indicates that the stud is difficult to remove from the surface. Meanwhile, PDMS has a low removal strength of 0.20 MPa , which shows the fouling release performance is excellent. The fouling release strength of HPCL-T0 is $\sim 0.40 \text{ MPa}$, much lower than that of the epoxy control. For HPCL-T12.5, HPCL-T25, HPCL-T37.5 and HPCL-T50, the strength values are 0.25, 0.20, 0.15, and 0.12 MPa , respectively. In other words, the coating containing PEG-PPG-PEG shows better fouling release ability than HPCL-T0. And the removal strength values of HPCL-Tx coatings are close to that of PDMS, which is well-known for its remarkable fouling release ability. As we know, an elastic surface is critical for the good fouling release of a coating [40]. Here, the coatings with PEG-PPG-PEG have notably lower modulus than HPCL-T0 (Fig. 1), leading to a better fouling release performance. And all the HPCL-Tx coatings have smooth surfaces (Fig. S2). Hence, the fouling does not adhere easily to the surface. Overall, low SEM and low surface roughness endow coatings with excellent fouling release performance.

The long-term antifouling property of HPCL-Tx coatings were examined in natural seawater. Fig. 8 shows the surfaces of HPCL-Tx after immersing in the South China Sea for 120 days. Before washing, the epoxy panel is completely covered by various marine fouling species including barnacles, tubeworms, encrusting bryozoans, and algae, which demonstrates the fouling pressure is severe. In contrast, relatively fewer fouling organisms are observed on the panel coated with HPCL-T0, because polymer degradation forms a renewing dynamic surface to reduce biofouling [28]. HPCL-T12.5 containing PEG-PPG-PEG exhibits better antifouling performance than HPCL-T0. However, as the PEG-PPG-PEG content further increases, the antifouling performance has declined. This differs from the data in laboratory bioassays, in which polymers with increasing content of amphiphilic additives show better antifouling efficacy. As discussed above, the amphiphilic additive can form a hydration layer on the surface, inhibiting bacterial adhesion. However, increasing PEG-PPG-PEG content lowers ester bond content, which reduces the degradation rate of coating and has a negative effect

on antifouling ability. Therefore, considering the synergy of degradation and fouling resistance, the HPCL-T25 with a suitable degradation rate and PEG-PPG-PEG content exhibits the best antifouling activity in natural seawater. After washing with a water jet, some fouling organisms are removed from coatings containing PEG-PPG-PEG, which indicates that the coatings have a good fouling release property. The result correlates well with the result of the pseudobarnacle removal test. This is because the coatings with PEG-PPG-PEG have an elastic and smooth surface that the fouling organisms are easily desorbed and removed. Overall, HPCL-T12.5 and HPCL-T25 without any biocide show excellent antifouling performance compared to the control.

The results above show that the multifunctional coating is capable of inhibiting marine biofouling. The mechanism is that the self-renewing dynamic surface with amphiphilic polymer inhibits the settlement of fouling organisms, and the elasticity of the coating weakens the adhesion strength of attached ones. As a result of the synergetic effect, the coating has long-term antifouling efficacy in the marine environment.

3. Conclusions

A degradable elastic antifouling material consisting of hyperbranched PCL and triblock copolymer PEG-PPG-PEG is developed. Such hybrid coating can degrade into small fragments to form a dynamic surface by the cleavage of hyperbranched PCL. The coating can resist the adsorption or adhesion of protein, bacteria *S. aureus* and *E. coli*, and diatom *N. incerta*, and the efficiency increases with the PEG-PPG-PEG content. Furthermore, the non-silicone coating has an elastic and smooth surface, which endows it with a good fouling release ability. Owing to these synergies, the polymeric coating free of biocide has an excellent performance in marine field test under heavy fouling pressure. This eco-friendly coating has potential for applications in marine anti-biofouling.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Author statement

The corresponding author is responsible for ensuring that the descriptions are accurate and agreed by all authors.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2022.107350>.

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